Bahareh Kaviani

Machine Learning Engineer | Deep Learning | Computer Vision

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Machine Learning Engineer specializing in Deep Learning and Computer Vision with 4+ years of experience developing AI solutions using Python and PyTorch. Experienced in building end-to-end machine learning pipelines, including data preparation, model development, training, evaluation, and optimization for real-world applications. Skilled in image classification, object detection, transfer learning, and deep learning model deployment. Strong background in robust deep learning and adversarial machine learning, with multiple IEEE publications in trustworthy AI and representation learning. Passionate about developing reliable, scalable, and production-oriented AI systems.

SKILLS

Programming Languages

- Python
- SQL/NoSQL
- MATLAB

Machine Learning

- Supervised & Unsupervised Learning
- Classification & Regression
- Clustering
- Feature Engineering
- Data Preprocessing
- Model Evaluation & Validation
- Hyperparameter Optimization
- Cross-Validation

Deep Learning

- PyTorch
- TensorFlow / Keras
- Convolutional Neural Networks (CNNs)
- Transfer Learning
- Representation Learning
- Generative Models (VAEs, GANs, Diffusion Models)
- Adversarial Machine Learning
- Model Optimization

Computer Vision

- Image Classification
- Object Detection
- YOLOv5 / YOLOv8 / YOLOv11
- R-CNN (R-CNN, Fast R-CNN, Faster R-CNN)
- Image Segmentation
- Image Preprocessing
- Data Augmentation
- Feature Extraction
- OpenCV
- Bounding Box Annotation
- IoU, NMS, Precision, Recall, mAP Evaluation

Data & Scientific Libraries

- NumPy
- Pandas
- Scikit-learn
- Matplotlib

Deployment & MLOps

- FastAPI
- Docker
- CI/CD Pipelines
- GitHub Actions

Development Tools

- Git
- Linux
- Jupyter Notebook
- VS Code

PROFESSIONAL EXPERIENCE

Graduate Research Assistant — Machine Learning & Computer Vision

University of Tehran — Tehran, Iran
Nov 2020 – Sep 2022

- Designed and implemented end-to-end deep learning pipelines for computer vision tasks, including data preprocessing, model development, training, evaluation, and robustness analysis using PyTorch.
- Developed CNN-based image classification models with transfer learning strategies using architectures such as ResNet and evaluated their performance under clean and adversarial settings.
- Built reproducible experimentation frameworks with modular dataset loaders, training loops, validation pipelines, checkpoint management, and quantitative model evaluation.
- Developed a VAE-based representation learning framework to improve the robustness and stability of deep neural networks against adversarial perturbations.
- Implemented and evaluated adversarial attack and defense pipelines including FGSM, R-FGSM, PGD, and MI-FGSM using white-box and black-box evaluation scenarios.
- Proposed the Separation Index (SI), a latent-space geometric metric for analyzing feature separability and robustness beyond traditional accuracy-based evaluation.

- Performed extensive experiments involving hyperparameter tuning, ablation studies, feature analysis, and comparison of multiple deep learning architectures.
- Contributed to IEEE publications in trustworthy AI, adversarial robustness, and robust representation learning.

Tech Stack: Python · PyTorch · CNNs · ResNet · VAEs · NumPy · Pandas · Matplotlib · Git · Linux

Undergraduate Research Assistant — Machine Learning & Computer Vision

University of Tehran — Tehran, Iran
Jul 2018 – Feb 2019

- Developed deep learning-based image classification systems for clothing recognition using CNN architectures and transfer learning techniques.
- Built complete computer vision pipelines including image preprocessing, augmentation, feature extraction, model training, and performance evaluation.
- Experimented with multiple CNN architectures including VGG-based models and compared different training strategies for improving classification accuracy.
- Designed a visual recommendation prototype using learned deep feature embeddings extracted from clothing images.
- Managed dataset preparation, image normalization, augmentation strategies, and evaluation workflows for large-scale image classification experiments.

Tech Stack: Python · TensorFlow · Keras · CNNs · Transfer Learning · NumPy · Matplotlib

Computer Vision Intern

School of Advanced Technologies in Medicine — Isfahan, Iran
Jul 2017 – Sep 2017

- Developed image processing pipelines for automated ocular image analysis using MATLAB.
- Implemented preprocessing, segmentation, feature extraction, and post-processing algorithms for OCT and retinal en-face images.
- Worked with medical imaging datasets and contributed to validation of computer vision algorithms in a clinical research environment.
- Contributed to an open-source MATLAB framework for automated ocular image analysis published in SoftwareX.

Tech Stack: Image Processing · Medical Imaging · Segmentation · Feature Extraction

SELECTED PROJECTS

Industrial PPE Detection System — YOLOv11

- Developed a real-time object detection system for industrial safety monitoring using YOLOv11.
- Built an end-to-end detection pipeline including dataset preparation, bounding box annotation, augmentation, training, validation, and inference.
- Evaluated detection performance using precision, recall, IoU, and mAP metrics.
- Implemented real-time inference on images and video streams using OpenCV.
- Optimized the model pipeline for practical deployment scenarios in industrial environments.

Skills: Python · PyTorch · YOLOv11 · Ultralytics · OpenCV · Computer Vision

Medical X-Ray Classification System — Deep Learning

- Developed a medical image classification pipeline for chest X-ray analysis using transfer learning with modern CNN architectures.
- Implemented custom PyTorch datasets, data preprocessing, augmentation, training, validation, and evaluation workflows.
- Addressed class imbalance and evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix.
- Designed a reproducible training framework suitable for real-world medical AI applications.

Skills: Python · PyTorch · CNNs · Transfer Learning · Medical Imagin

Production ML Deployment Pipeline

- Developed an end-to-end machine learning deployment pipeline for serving deep learning models through REST APIs.
- Built inference APIs using FastAPI with model loading, prediction endpoints, and health monitoring.
- Containerized applications using Docker and implemented automated testing workflows using GitHub Actions.
- Integrated engineering practices include modular code structure, testing, logging, and reproducible environments.
- **Skills:** PyTorch · FastAPI · Docker · REST API · GitHub Actions · Pytest

Real-Time Object Detection and Tracking

- Developed computer vision pipelines for real-time object localization and tracking from webcam/video streams.
- Implemented image processing techniques, feature extraction, and motion analysis algorithms.
- Extracted object position and movement information for real-time monitoring applications.
- Optimized processing pipelines for low-latency inference.
- **Skills:** Python · PyTorch · OpenCV

EDUCATION

University of Tehran | 2019 – 2022

- **M.Sc. in Electrical Engineering (Control Engineering) |** GPA: 17.19 / 20
- Thesis: *Defending Deep Neural Networks Against Adversarial Attacks*

University of Tehran | 2014 – 2019

- **B.Sc. in Electrical Engineering (Control Engineering) |** GPA: 16.01 / 20
- Thesis: *Deep Learning-Based Clothing Image Classification using CNNs and Transfer Learning*

SELECTED PUBLICATIONS

Density Estimation Helps Adversarial Robustness

IEEE ICCKE, 2023

- Proposed a density-aware approach for improving adversarial robustness through enhanced latent feature representation and evaluation.

Adversarial Robustness Evaluation with Separation Index

IEEE ICCKE, 2023

- *Proposed the Separation Index (SI), a novel geometric metric for evaluating adversarial robustness beyond conventional accuracy-based measures.*

Automatic Multifaceted MATLAB Package for Ocular Image Analysis

***SoftwareX**, 2019*

- *Contributed to the development of an open-source MATLAB package for automated ocular image analysis, including image preprocessing, segmentation, and feature extraction.*

TEACHING EXPERIENCE

University of Tehran | 2017 – 2022

- Assisted in teaching graduate and undergraduate courses in Deep Learning, Statistical Inference, System Identification, and Machine Learning, including laboratory sessions, programming assignments, and grading.

LANGUAGES

- English: Professional Working Proficiency (TOEFL iBT: 85)
- Persian – Native